

PulmoDataSets: An R Package for Teaching Applied Statistics and Data Science in Pulmonary Medicine

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Summary

PulmoDataSets is an R package that provides a comprehensive and curated collection of datasets related to the lungs, the respiratory system, and associated diseases (Caceres Rossi, 2025a). The package includes epidemiological, clinical, experimental, and simulated datasets covering conditions such as lung cancer, asthma, Chronic Obstructive Pulmonary Disease (COPD), tuberculosis, whooping cough, pneumonia, influenza, and other respiratory illnesses. Designed as open-source educational technology, it supports the teaching of applied statistics and data science in the context of pulmonary medicine, making curated respiratory health data immediately accessible to students, educators, data scientists, physicians, and pulmonary specialists in a single, ready-to-use R package.

Statement of Need

R is a programming language with a rich and well-established culture of dataset packages (R Core Team, 2025; Wickham & Bryan, 2023). However, the majority of these packages contain datasets that are dispersed across multiple topics and domains. Widely used packages such as MASS (Venables & Ripley, 2002), datasets (R Core Team, 2025), and openintro (Cetinkaya-Rundel et al., 2024) provide heterogeneous collections that span diverse subject areas without thematic focus. Specialized dataset packages curating a single, cohesive topic remain scarce in the R ecosystem, and packages focused specifically on medicine or any of its branches are even rarer.

Respiratory diseases represent one of the leading causes of morbidity and mortality worldwide (Ferkol & Schraufnagel, 2014; Labaki & Han, 2020). Despite the clinical and epidemiological relevance of pulmonary medicine, no dataset package in R is specifically dedicated to the lungs and respiratory diseases. Instructors and students working in biostatistics, epidemiology, data science, and pulmonary medicine are frequently required to search across multiple sources, clean, and restructure data before any meaningful statistical analysis or classroom exercise can begin.

PulmoDataSets addresses this gap by providing a curated, ready-to-use collection of datasets exclusively focused on the lungs, the respiratory system, and associated diseases (Caceres Rossi, 2025a). The package introduces two design features that distinguish it from existing dataset packages in R. First, each dataset uses a suffix to explicitly identify its R object type: `_df` for data frames, `_dt` for data tables, `_tbl_df` for tibbles, and `_ts` for time series objects. This naming convention allows users to immediately identify the structure of a dataset before loading it, reducing friction in both teaching and analytical workflows. Second, the package includes the utility function `view_datasets_PulmoDataSets()`, which prints a complete, formatted list of all available datasets directly in the R console — a feature absent in most dataset packages in the R ecosystem.

41 Table 1 summarizes the key differences between PulmoDataSets and three widely used
42 dataset packages in R.

Table 1: Comparison of PulmoDataSets with selected dataset packages in R.

Feature	datasets	MASS	openintro	PulmoDataSets
Thematic focus	No	No	No	Yes
Pulmonary medicine datasets	No	No	No	Yes
Object type suffixes (<code>_df</code> , <code>_dt</code> , <code>_tbl_df</code> , <code>_ts</code>)	No	No	No	Yes
Dataset listing function	No	No	No	Yes
Available on CRAN	Yes	Yes	Yes	Yes
Active GitHub repository	N/A	No	Yes	Yes

43 Functionality

44 PulmoDataSets is available on CRAN and can be installed and loaded as follows:

```
install.packages("PulmoDataSets")
library(PulmoDataSets)
```

45 The package contains 36 datasets covering the lungs, the respiratory system, and associated
46 diseases. The datasets span a wide range of topics including lung cancer, asthma, Chronic
47 Obstructive Pulmonary Disease (COPD), tuberculosis, whooping cough, pneumonia,
48 influenza, smoking, and air pollution. They include epidemiological, clinical, experimental,
49 simulated, and historical data, making PulmoDataSets suitable for a broad range of
50 statistical and data science applications in pulmonary medicine.

51 The design of PulmoDataSets follows the Rossi Design Philosophy (RDP) (Caceres Rossi,
52 2025b), a systematic methodology for developing R dataset packages that enhances
53 usability, transparency, and discoverability. The RDP is built on four core principles:

- 54 • **Structural Transparency:** dataset names use standardized suffixes to indicate
55 their R object type.
- 56 • **Predictive Consistency:** a uniform naming pattern (`dataset_name_suffix`) ensures
57 predictability across all datasets.
- 58 • **Enhanced Discoverability:** a utility function lists all available datasets directly
59 in the R console.
- 60 • **User-Centric Documentation:** object types are explicitly stated in dataset names
61 and documentation.

62 In practice, each dataset in PulmoDataSets uses one of four suffixes to identify its R object
63 type:

- 64 • `_df`: standard R data frame
- 65 • `_dt`: data table (`data.table` ecosystem)
- 66 • `_tbl_df`: tibble (`tidyverse` ecosystem)
- 67 • `_ts`: time series (`ts` object, base R)

68 For example, `bronchitis_Cardiff_df` is a data frame, `asthma_patients_tbl_df` is a tibble,
69 `ai_ipn_performance_dt` is a data table, and `pneumonia_influenza_ts` is a time series object.
70 This naming convention allows users to immediately identify the structure of a dataset
71 before loading it, reducing friction in both teaching and analytical workflows. Furthermore,
72 each dataset includes a documentation tooltip that explicitly states its R object type
73 alongside a description of its contents, providing an additional reference point for users
74 unfamiliar with the suffix convention.

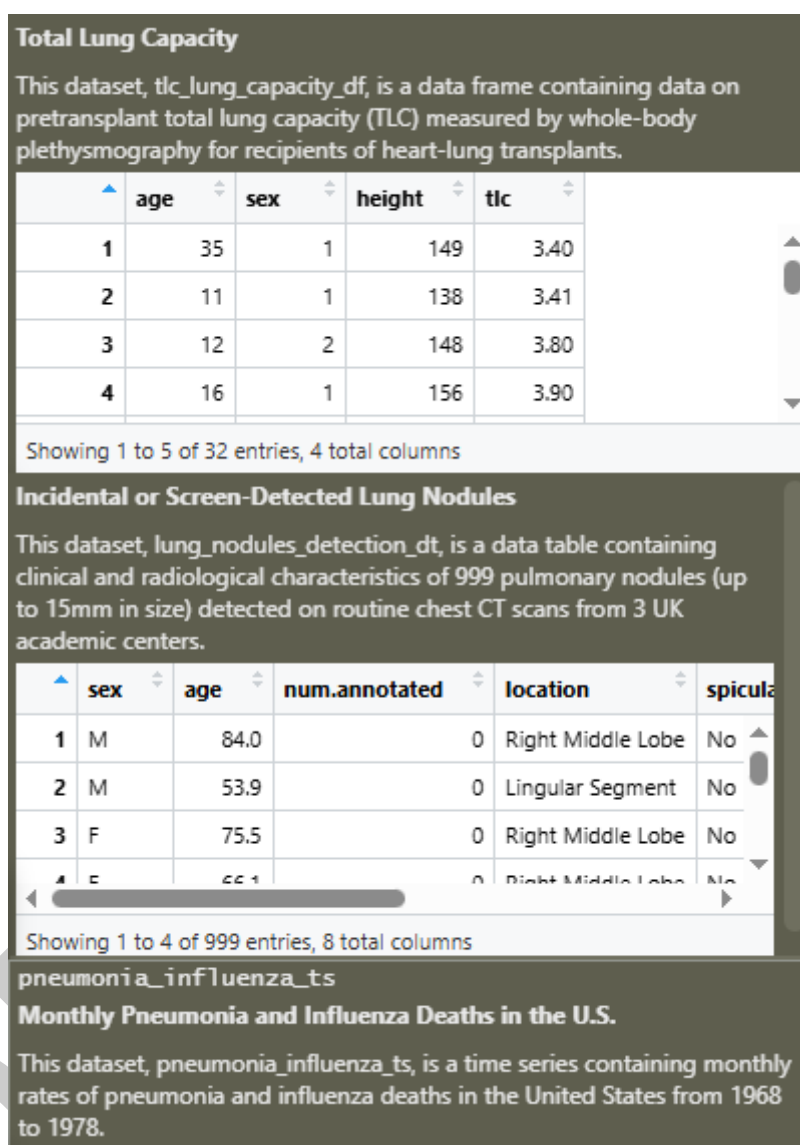


Figure 1: Dataset tooltips in RStudio showing the R object type explicitly stated in the documentation for `tlc_lung_capacity_df` (data frame), `lung_nodules_detection_dt` (data table), and `pneumonia_influenza_ts` (time series).

75 The package also includes the utility function `view_datasets_PulmoDataSets()`, which
76 prints a complete, formatted list of all available datasets directly in the R console:

```
library(PulmoDataSets)
```

```
view_datasets_PulmoDataSets()
```

77 Datasets available in the 'PulmoDataSets' package:

```
78 [1] "UK_female_lung_deaths_ts"      "UK_male_lung_deaths_ts"
79 [3] "USMortality_df"               "USRegionalMortality_df"
80 [5] "ai_ipn_performance_dt"        "air_pollution_mortality_df"
81 [7] "asthma_patients_tbl_df"       "bronchitis_Cardiff_df"
82 [9] "chicago_pollution_df"       "child_wheeze_pollution_df"
83 [11] "children_respiratory_rates_df" "danish_lung_incidence_df"
84 [13] "engwales_cancer_mortality_df" "influenza_us_1975_df"
```

```

85 [15] "lung_cancer_survival_df"      "lung_nodules_detection_dt"
86 [17] "lungca_cancer_deaths_df"     "neonatal_intubation_times_df"
87 [19] "nicotine_gum_df"             "ohio_children_wheeze_df"
88 [21] "patients_lung_diseases_tbl_df" "pneumonia_influenza_ts"
89 [23] "respiratory_clinical_trial_df" "respiratory_infections_df"
90 [25] "respiratory_trial_df"         "respiratory_trial_outcomes_df"
91 [27] "smoking_UK_tbl_df"           "smoking_doctors_df"
92 [29] "smoking_lung_cancer_df"       "smoking_youth_tbl_df"
93 [31] "tlc_lung_capacity_df"         "tuberculosis_vaccine_df"
94 [33] "veterans_lung_cancer_df"      "whooping_cough_dk_df"
95 [35] "whooping_cough_phila_df"      "whooping_cough_ts"

96 Individual datasets can be loaded and explored using standard R functions:

# Load a dataset
data(nicotine_gum_df)

# Preview the first rows
head(nicotine_gum_df)

# Open in RStudio Viewer
View(nicotine_gum_df)

```

Usage in Teaching and Learning

PulmoDataSets is designed to support both instructors and students in teaching and learning applied statistics and data science in the context of pulmonary medicine.

From the instructor's perspective, PulmoDataSets eliminates the time-consuming process of searching, cleaning, and preprocessing data before it can be used in a classroom setting. With 36 curated, ready-to-use datasets, instructors can immediately design exercises, assignments, and demonstrations covering a wide range of statistical methods. The diversity of R object types available in the package — data frames, tibbles, data tables, and time series objects — allows instructors to select datasets aligned with the specific R paradigm taught in their course, whether base R, the tidyverse ecosystem, the `data.table` framework, or time series analysis. For example, datasets such as `bronchitis_Cardiff_df` and `lung_cancer_survival_df` are well-suited for logistic regression and survival analysis exercises, while `pneumonia_influenza_ts` and `whooping_cough_ts` support time series decomposition and forecasting demonstrations. Datasets such as `tuberculosis_vaccine_df` and `respiratory_clinical_trial_df` can be used to illustrate clinical trial methodology and vaccine efficacy analysis.

From the student's perspective, PulmoDataSets provides immediate access to publicly available pulmonary-related datasets from the first session in R. The utility function `view_datasets_PulmoDataSets()` allows students to explore the full catalog of available datasets directly in the R console, without consulting external documentation. The standardized suffix naming convention — a core principle of the Rossi Design Philosophy (Caceres Rossi, 2025b) — enables students to identify the structure of each dataset before loading it, reducing confusion and accelerating the learning process. Working with thematically focused, domain-specific datasets strengthens the connection between statistical methods and their practical applications in pulmonary medicine and public health.

PulmoDataSets is suitable for adoption in undergraduate and graduate courses in biostatistics, epidemiology, data science, and public health, as well as in research training programs and self-directed learning in pulmonary medicine (Doudesis & Manataki, 2022).

Story of the Project

In mid-2024, I developed my first R package, `timeSeriesDataSets` (Caceres Rossi, 2024), a specialized collection of datasets designed for teaching and working with time series analysis in R. Through the development of that project, I became aware of a broader structural gap within the R ecosystem: while many general-purpose dataset packages exist, there is a limited number of thematically specialized dataset collections tailored to specific scientific domains.

Conversations with colleagues — particularly pulmonologists, medical researchers, and university professors — reinforced this observation. Many expressed the difficulty of finding curated, analysis-ready respiratory datasets suitable for research, classroom demonstrations, methodological testing, and statistical instruction. A significant amount of time was often spent locating, cleaning, harmonizing, and structuring data before it could be used pedagogically or analytically.

`PulmoDataSets` emerged as a response to this need (Caceres Rossi, 2025a).

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